

NUKUS, Uzbekistan

WMO: 382640

Lat: 42.483N

Lon: 59.617E

Elev: 77

StdP: 100.40

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-16.2	-23.4	0.5	-17.1	-20.2	0.6	-13.9	10.2	0.2	9.1	-0.7	3.1	10	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	39.8	21.6	38.1	20.9	36.9	20.6	24.6	35.5	23.5	34.1	22.4	33.0	4.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.0	30.9	19.9	14.8	29.9	18.7	13.7	29.1	75.2	35.5	70.5	33.9	66.4	32.9	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.0	7.9	-21.2	42.4	4.3	1.5	-24.2	43.5	-26.7	44.4	-29.1	45.3	-32.2	46.4		
(5)			-21.4	25.5	4.2	2.2	-24.4	27.0	-26.9	28.3	-29.2	29.5	-32.3	31.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	-3.7	-1.2	6.9	15.0	22.2	27.1	28.9	26.9	20.2	12.4	4.2	-1.6	(6)	
	DBStd	12.49	6.29	6.27	5.59	5.21	3.81	3.21	2.88	3.40	3.99	4.93	5.42	5.37	(7)	
	HDD10.0	1462	426	316	128	15	0	0	0	0	1	32	184	361	(8)	
	HDD18.3	2983	684	547	355	124	12	1	0	1	26	191	424	618	(9)	
	CDD10.0	2621	0	2	32	166	377	512	585	525	305	107	10	0	(10)	
	CDD18.3	1100	0	0	1	25	130	263	327	267	81	7	0	0	(11)	
	CDH23.3	15362	0	1	23	322	1641	3727	4803	3686	1044	115	1	0	(12)	
	CDH26.7	8590	0	0	4	109	753	2169	2931	2164	438	21	0	0	(13)	
(14) Wind	WSAvg	3.7	3.5	4.0	4.3	4.3	4.0	3.9	3.8	3.8	3.4	3.1	3.2	3.4	(14)	
(15) Precipitation	PrecAvg	106	10	8	16	19	14	3	3	1	2	6	8	13	(15)	
	PrecMax	238	24	23	57	78	53	29	18	14	14	31	52	30	(16)	
	PrecMin	37	0	0	1	1	0	0	0	0	0	0	0	0	(17)	
	PrecStd	50	7	7	15	18	15	6	5	3	4	8	11	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	17.1	26.7	32.9	37.0	40.9	42.2	41.5	37.1	29.0	21.1	13.1	(19)	
		MCWB	6.4	9.8	14.1	17.9	18.8	21.0	21.3	23.4	20.9	16.2	12.6	7.7	(20)	
	2%	DB	7.9	12.5	21.9	30.0	34.7	38.9	40.1	39.0	33.2	26.5	17.2	9.8	(21)	
		MCWB	5.0	7.0	12.2	15.7	17.6	20.7	21.8	21.7	19.0	14.4	10.2	6.2	(22)	
	5%	DB	5.6	9.2	18.8	27.0	32.9	37.1	38.3	37.2	31.1	24.0	15.0	7.2	(23)	
		MCWB	3.4	4.9	10.6	14.7	17.4	20.2	21.2	21.0	17.5	13.5	9.0	4.5	(24)	
	10%	DB	3.7	6.6	15.9	24.4	31.0	35.7	36.9	35.8	29.1	21.6	12.2	5.2	(25)	
		MCWB	2.1	3.5	9.0	13.7	16.7	19.9	20.9	20.5	16.5	12.4	7.5	3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.8	14.9	19.3	21.8	25.2	26.4	25.9	22.6	17.8	14.0	8.5	(27)	
		MCDB	9.5	15.9	23.9	29.5	32.3	35.7	36.4	37.8	33.1	26.6	19.6	11.8	(28)	
	2%	WB	5.1	7.5	12.9	17.3	20.0	23.7	24.9	24.3	20.6	15.6	11.5	6.5	(29)	
		MCDB	7.5	11.5	20.4	26.8	30.2	34.0	35.4	35.9	30.3	24.0	16.4	9.2	(30)	
	5%	WB	3.6	5.3	11.2	15.9	18.7	22.5	23.6	23.0	19.2	14.2	9.7	4.7	(31)	
		MCDB	5.3	8.5	17.6	24.6	28.6	33.0	33.9	34.2	28.8	22.1	13.4	6.8	(32)	
	10%	WB	2.1	3.6	9.6	14.5	17.7	21.3	22.6	21.8	17.7	13.0	7.9	3.2	(33)	
		MCDB	3.5	6.2	15.2	22.7	27.9	31.9	33.0	32.6	26.8	20.0	11.5	5.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	9.4	11.0	12.2	13.4	13.9	13.6	14.6	14.7	13.4	10.3	8.1	(35)	
		MCDBR	9.2	12.8	14.8	15.4	16.2	16.1	15.6	16.5	17.0	16.5	14.2	10.9	(36)	
	5% WB	MCWBR	6.5	7.7	7.8	6.8	6.2	5.9	5.6	6.1	7.6	8.3	8.2	7.7	(37)	
		MCWBR	8.5	11.6	13.5	13.4	13.3	13.7	13.5	14.8	15.1	14.2	11.5	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.383	0.413	0.474	0.530	0.536	0.525	0.509	0.481	0.450	0.456	0.418	0.386	(40)	
		taud	2.049	1.988	1.895	1.800	1.829	1.880	1.943	1.989	2.015	1.954	2.016	2.047	(41)	
	Ebn at Noon		706	753	755	746	755	765	772	780	771	695	656	657	(42)	
		Edh at Noon	112	140	174	206	206	196	182	169	152	143	116	104	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.58	2.60	3.99	5.40	6.67	7.50	7.27	6.71	5.31	3.53	2.12	1.44	(44)	
	RadStd		0.19	0.28	0.34	0.36	0.38	0.33	0.24	0.20	0.20	0.21	0.25	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	+0.69	-2.24	-2.79	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)	N/A	N/A	N/A	+0.83	N/A	-0.67	N/A	N/A	N/A	+52	

Nomenclature: See separate page